

Contour Integration

Application of Residues Theorem

1. $\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta$, C is a circle with unit radius, where F is a rational function.

$$z = e^{i\theta}$$

$$dz = i e^{i\theta} d\theta = iz d\theta$$

$$d\theta = \frac{dz}{iz}$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + z^{-1}}{2}$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{z - z^{-1}}{2i}$$

$$\int_C F\left(\frac{z + z^{-1}}{2}, \frac{z - z^{-1}}{2i}\right) \frac{dz}{iz}$$

$$= 2\pi i (\text{Sum of residues of } f(z) \text{ at the poles lying inside } C.)$$

Q. Evaluate the following integration by contour integration:

$$\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta$$

Solⁿ:

We have, $e^{i3\theta} = \cos 3\theta + i \sin 3\theta$

Let, $z = e^{i\theta} \Rightarrow z^3 = e^{i3\theta}$

$$\begin{aligned} dz &= i e^{i\theta} d\theta \Rightarrow d\theta = \frac{dz}{iz} \\ &= iz d\theta \end{aligned}$$

$$\cos \theta = \frac{z + z^{-1}}{2} = \frac{z + 1/z}{2} = \frac{z^2 + 1}{2z}$$

R.P. of

$$\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta = \text{R.P. of } \int_0^{2\pi} \frac{e^{i3\theta}}{5 - 4 \cos \theta} d\theta$$

$$= \text{R.P. of } \int_C \frac{z^3}{5 - 4 \left(\frac{z^2 + 1}{2z} \right)} \cdot \frac{dz}{iz}$$

$$= \text{R.P. of } \int_C \frac{z^3 (2z)}{10z - 4z^2 - 4} \cdot \frac{dz}{i\cancel{z}}$$

$$= \text{R.P. of } \int_C \frac{2z^3}{i(10z - 4z^2 - 4)} dz$$

$$= \text{R.P. of } \frac{1}{i} \int_C \frac{2z^3}{-(4z^2 - 10z + 4)} dz$$

$$= \text{R.P. of } \cancel{i} \int_C \cancel{-} \frac{2z^3}{(4z^2 - 10z + 4)} dz$$

$$\left[\frac{1}{i} = \frac{1}{i} \times \frac{i}{i} = \frac{i}{i^2} = \frac{i}{-1} = -i \right]$$

$$= \text{R.P. of } i \int_C \frac{2z^3}{4z^2 - 10z + 4} dz$$

$$\begin{aligned} \text{Let, } f(z) &= \frac{2z^3}{4z^2 - 10z + 4} = \frac{\cancel{2}z^3}{\cancel{2}(2z^2 - 5z + 2)} \\ &= \frac{z^3}{(z-2)(2z-1)} \end{aligned}$$

For poles of $f(z)$,

$$(z-2)(2z-1) = 0$$

$$\Rightarrow z = 2, \frac{1}{2}$$

$z = \frac{1}{2}$ lies inside circle C , $|z| = 1$

$z = \frac{1}{2}$ is simple pole.

$$f(z) = \frac{g(z)}{h(z)}$$

Here, $g(z) = z^3 \Rightarrow g(1/2) = 1/8 \neq 0$
&

$$h(z) = (z-2)(2z-1) \Rightarrow h(1/2) = 0$$

So, $h'(z) = 2(z-2) + (2z-1)$

$$\Rightarrow h'(1/2) = 2\left(\frac{1}{2} - 2\right) + \left(2 \cdot \frac{1}{2} - 1\right)$$

$$= 2\left(\frac{-3}{2}\right) + 0$$

$$= -3$$

$$\therefore \text{Res}(z=1/2) = \frac{g(1/2)}{h'(1/2)} = \frac{1/8}{-3} = -\frac{1}{24}$$

Now,

$$\int_C \frac{z^3}{2z^2 - 5z + 2} dz = 2\pi i \left(\frac{-1}{24} \right) = -\frac{\pi i}{12}$$

Thus, $\int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos\theta} d\theta = \text{R.P. of } \int_0^{2\pi} \frac{e^{3i\theta}}{5-4\cos\theta} d\theta$
 $= \text{R.P. of } i \int \frac{z^3}{2z^2 - 5z + 2} dz$

$$= \text{R.P. of } i \left(\frac{-\pi i}{12} \right) = \frac{\pi}{12}$$

$$= \text{R.P. of } \left(\frac{\pi}{12} \right)$$

$$\therefore \int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta = \frac{\pi}{12}$$

Q. Evaluate $\int_0^{2\pi} \frac{d\theta}{a + b \sin \theta}$, $a > |b|$.

Solⁿ:

Let, $z = e^{i\theta} \Rightarrow dz = i e^{i\theta} d\theta \Rightarrow dz = iz d\theta$

$$d\theta = \frac{dz}{iz}$$

$$\sin \theta = \frac{z - z^{-1}}{2i} = \frac{z - 1/z}{2i} = \frac{z^2 - 1}{2iz}$$

$$\int_0^{2\pi} \frac{d\theta}{a + b \sin \theta} = \int_C \frac{1}{a + b \left(\frac{z^2 - 1}{2iz} \right)} \cdot \frac{dz}{iz}$$

$$= \int_C \frac{2iz}{2iaz + bz^2 - b} \cdot \frac{dz}{iz}$$

$$= 2 \int_C \frac{dz}{bz^2 + 2iaz - b}$$

$$= \frac{2}{b} \int_C \frac{dz}{z^2 + \frac{2iaz}{b} - 1}$$

$$f(z) = \frac{1}{z^2 + \frac{2iaz}{b} - 1}$$

For poles of $f(z)$,

$$z^2 + \frac{2iaz}{b} - 1 = 0$$

$$z = \frac{-\frac{2ai}{b} \pm \sqrt{\frac{4i^2a^2}{b^2} + 4}}{2}$$

$$\Rightarrow z = \frac{-\frac{2ai}{b} \pm \frac{2}{b} \sqrt{-a^2 + b^2}}{2}$$

$$\Rightarrow z = -\frac{ai}{b} \pm \frac{\sqrt{-a^2 + b^2}}{b}$$

$$\Rightarrow z = -\frac{ai}{b} \pm \frac{\sqrt{-a^2 + b^2}}{b}$$

$$\text{Let, } \alpha = \frac{-ai}{b} + \frac{\sqrt{-a^2 + b^2}}{b} = -\left(\frac{ai}{b} - \frac{\sqrt{-a^2 + b^2}}{b}\right)$$

$$\& \beta = \frac{-ai}{b} - \frac{\sqrt{-a^2 + b^2}}{b} = -\left(\frac{ai}{b} + \frac{\sqrt{-a^2 + b^2}}{b}\right)$$

$$\text{Given, } a > |b| \Rightarrow \left|\frac{a}{b}\right| > 1$$

$$\text{So, } |\alpha| < 1 \& |\beta| > 1$$

$$\text{So, } z = \alpha \text{ lies inside } C \quad |z| = 1.$$

$\therefore z = \alpha$ is a simple pole.

$$\therefore \text{Res}(z = \alpha) = \lim_{z \rightarrow \alpha} \left[\cancel{(z - \alpha)} \frac{1}{\cancel{(z - \alpha)}(z - \beta)} \right]$$

$$= \lim_{z \rightarrow \alpha} \left(\frac{1}{z - \beta} \right)$$

$$= \frac{1}{\alpha - \beta}$$

Now,

$$\alpha - \beta = \cancel{\frac{-ai}{b}} + \frac{\sqrt{-a^2 + b^2}}{b} + \cancel{\frac{ai}{b}} + \frac{\sqrt{-a^2 + b^2}}{b}$$

$$\Rightarrow \alpha - \beta = \frac{2\sqrt{-a^2 + b^2}}{b}$$

$$\text{So, Res}(z = \alpha) = \frac{b}{2\sqrt{-a^2 + b^2}}$$

$$\text{Then, } \int_C \frac{dz}{(z - \alpha)(z - \beta)} = 2\pi i \left(\frac{b}{2\sqrt{-a^2 + b^2}} \right)$$

$$\therefore \int_0^{2\pi} \frac{d\theta}{a + b \sin \theta} = \frac{2}{b} \left[\frac{\pi i b}{\sqrt{-a^2 + b^2}} \right] = \frac{2\pi i}{\sqrt{-a^2 + b^2}}$$

$$= \frac{2\pi}{\sqrt{a^2 - b^2}}$$

$$\left[\sqrt{-a^2 + b^2} = \sqrt{-(a^2 - b^2)} = \sqrt{(-1)} \cdot \sqrt{a^2 - b^2} = \sqrt{i^2} \sqrt{a^2 - b^2} = i\sqrt{a^2 - b^2} \right]$$